8 A sinusoidal alternating input voltage V_{IN} is supplied to a circuit that produces a rectified and smoothed output voltage V_{OUT}.

The variations with time t of V_{IN} and V_{OUT} are shown in Figure 8.1 and Figure 8.2 respectively.

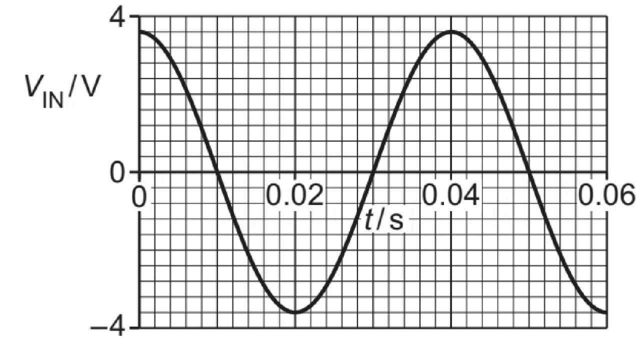

Figure 8.1

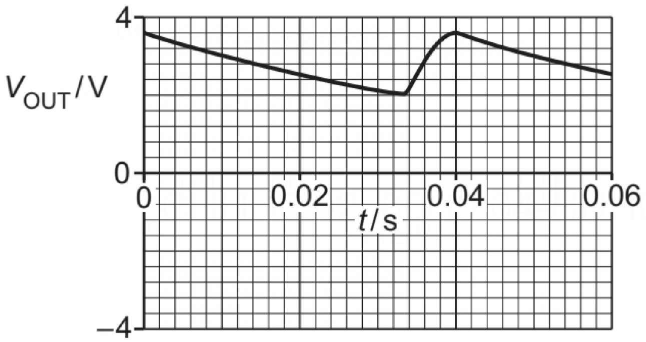

Figure 8.2

(a) **(i)** State the name of the component in the circuit that produces the rectification.

.... [1]

(ii) State the name of the component in the circuit that produces the smoothing.

.... [1]

(b) **(i)** Determine the peak value of V_{IN}.

peak value of V_{IN} = ... V [1]

(ii) Determine the period of V_{IN}.

period of V_{IN} = ... s [1]

(c) Apart from the peak value and period of V_{IN}, determine **three** other conclusions that can be drawn from Figure 8.1 and Figure 8.2 about the input and output voltages and about the circuit. The conclusions may be qualitative or quantitative. Use the space for any working.

1 ....

....

2 ....

....

3 ....

....

[3]

[Total: 7]